

Java FSD - Day 3 to Day 4 Summary

Day 3 - Java Basics

- Java is a case-sensitive programming language.
- Every statement ends with a semicolon (;).
- Curly braces {} are used to define blocks of code.
- Class names should begin with uppercase letters.
- Method names and variable names should begin with lowercase letters.
- The file name must be the same as the public class name.
- Single-line comments are written using //.
- Multi-line comments are written using /* ... */.
- Data types are divided into Primitive and Non-Primitive types.

Day 4 - Variables, Type Casting, and Operators

- Variables are used to store data values.
- Type Casting is the process of converting one data type into another.
- Widening Type Casting (Implicit): Small to large data types, automatic conversion, and no data loss.
- Narrowing Type Casting (Explicit): Large to small data types, manual conversion, and possible data loss.
- Arithmetic Operators: +, -, *, /, %
- Assignment Operators: =, +=, -=, *=, /=
- Relational Operators: ==, !=, >, <, >=, <=
- Logical Operators: &&, ||, !
- Unary Operators: ++, --
- Bitwise Operators: &, |, ^, ~, <<, >>

Practice Programs

- Addition of Two Numbers
- Simple Interest Calculator
- Area of Rectangle
- Celsius to Fahrenheit Converter
- Perimeter of a Square

Handwritten Notes

Original classroom notes are attached below for reference.

DAY-3

2025
JANUARY

Monday 13

Java Syntax Rules:

- ↳ Every statement end with (;)
- ↳ { } define block of code
- ↳ class name only with upper case
- ↳ Method & Variable name start lower case
- ↳ Java is case-sensitive
- ↳ The file name must match the public class name.

Comments in JAVA

- (i) Single-line comment //
- (ii) Multiple-line comment /* Adams Hello */

Data Type

- ↳ To allocate proper memory for variables
- ↳ (i) Primitive Data Type
- (ii) Non-Primitive Data Type



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DAY - 4

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Wednesday 15

Variables:

Type Casting:

- ↳ Convert one data type into another data type
- ↳ To store a value of one data type into a variable of another data type.

↳ Use of Type Casting,

- Perform operation b/w datatype
- Convert smaller datatype to larger ones.

Implicit
(Auto) Widening - Explicit
(Manual) Narrowing - Implicit
Explicit

Types,

(i) Widening Type Casting:

- Implicit type casting (or) Upcasting
- Done automatically.
- No data loss.
- Order of data types [Small - large]

(ii) Narrowing Type Casting:

- Explicit type casting (or) downcasting
- Done Manually,
- Leads to data loss



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Rules for Type Casting:

(i) Widening

- Smaller $\rightarrow$ Larger
- No data loss
- Safe conversion

(ii) Narrow

- Larger $\rightarrow$ Smaller
- Yes, data loss is there

Operators in Java= to assign value $[A+B=10]$ == to check both the side $[A==B]$

& multiply

(ii) OR addition

(i) Arithmetic Operators, $(+, -, \%, /, *)$ (ii) Assignment Operators, $(+=, -=, *=, /=)$ (iii) Relational Operators, $(==, !=, >, <, >=, <=)$ (iv) Logical Operators, $(\&\&, \|\|, ==)$ Ex: $(!(a==10))$

(v) Unary operators

(vi) Bitwise Operators,



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